

Cutoff scaling for the normal moment integral

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Abstract

The paper's bare normal moment integral lives on $[0, b]^{|I|}$ with an arbitrary cutoff $b > 0$; the monomial milestones were stated on the unit box. The substitution $u = bv$ on $\mathbb{R}^d$ gives the exact identity $M_b(N) = b^{\sum_i (h_i+1)} M_1(N b^{2\sum_i k_i})$ (`monomialBoxRealCutoff_eq`), and hence the mixed-ratio asymptotic on $(0, b]^d$ with the extra factor $b^{\sum_i (h_i+1) - 2\lambda \sum_i k_i} = \prod_{i \notin J} b^{h_i+1-2k_i\lambda}$ — exactly the b -dependence of the paper's Laurent coefficient $a_{-|J|}$ — and no b -dependence at all in the equal-ratio case. Zero `sorry/axiom`.

1 The things formalised

```
def cutoffBox (d : ℕ) (b : ℝ) : Set (Fin d → ℝ) := Set.pi univ fun _ => Ioc 0 b
theorem cutoffBox_eq_smul (d b) (hb : 0 < b) : cutoffBox d b = b • unitBox d
def monomialBoxRealCutoff (d) (h k) (b N : ℝ) : ℝ :=
  x in cutoffBox d b, ( i, x i ^ h i) * exp (-( i * N * i, x i ^ (2 * k i)))
theorem monomialBoxRealCutoff_eq (d h k b N) (hb : 0 < b) :
  monomialBoxRealCutoff d h k b N
  = b ^ ( i, (h i + 1)) * monomialBoxReal d h k (N * b ^ (2 * i, k i))
theorem cutoff_power_eq (d h k b l) (hb) :
  b ^ ((( i, (h i + 1)) : ℕ) : ℕ) - 2 * l * ( i, k i)
  = b ^ ( i, (h i + 1)) * (b ^ (2 * i, k i)) ^ (-l)
theorem tendsto_log_scale_pow (c) (hc : 0 < c) (r) :
  Tendsto (fun N => (log (N * c) / log N) ^ r) atTop ( 1)
theorem monomialBoxRealCutoff_mixed_tendsto / _isEquivalent ... :
  (fun N => monomialBoxRealCutoff (d + 1) h k b N) ~[atTop]
  fun N => b ^ (( i, (h i + 1)) - 2 * l * i, k i) * monomialMixedConst h k l
  * N ^ (-l) * log N ^ (multCount (ratioExp h k) l - 1)
theorem cutoff_power_eq_one_of_equal (hratio : i, ratioExp h k i = 1) :
  (( i, (h i + 1)) : ℕ) : ℕ) - 2 * l * ( i, k i) = 0
```

2 Mathematics

$(0, b]^d = b \cdot (0, 1]^d$ and Mathlib's `setIntegral_comp_smul_of_pos` on the d -dimensional space give $\int_{b \cdot S} f = b^d \int_S f(bv) dv$; the monomial factor contributes $b^{\sum h_i}$ and the exponent absorbs $b^{2\sum k_i}$ into N . The asymptotic transfer is a product of two limits: the unit-box ratio along $N \mapsto N b^{2\sum k_i}$ and $(\log(Nc)/\log N)^r \rightarrow 1$; the scale factors combine as $b^{\sum (h_i+1)} c^{-\lambda} = b^{\sum (h_i+1) - 2\lambda \sum k_i}$.

3 Paper-facing meaning

Item (iii) of Astra #15's list: the cutoff dependence of the leading coefficient matches eq. `a_minus_m_explicit` ($\prod_{i \notin J} b^{h_i+1-2k_i\lambda}$), and is absent when all ratios are equal. Astra #16

(received during this unit) ranks the continuous-amplitude milestone next and records a leading-coefficient qualification for mixed ratios (the amplitude is integrated over the nonminimal normal directions, not evaluated at the origin).

4 Lean notes

- Pointwise set scaling `b`
- `S` needs `open scoped Pointwise`.
- `setIntegral_comp_smul_of_pos` lives in `MeasureTheory.Measure`; the dimension is `Module.finrank_fin_fun`.
- The asymptotic with cutoff is *not* an eventual equality with the unit-box ratio (the log factors differ); prove it as a product of limits and close the algebra after `div_pow` with explicit nonvanishing of both log powers.
- Again: a local named `h` shadows the exponent function; use `hT`.